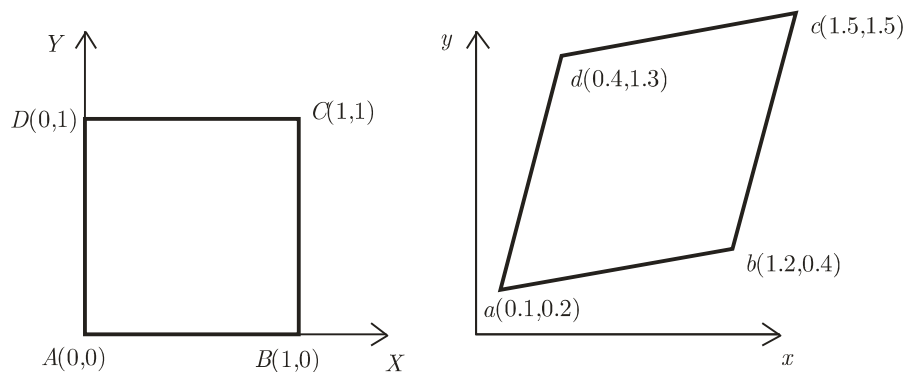


Examen Parcial 24/06/2013

1. Sea el cuadrado ABCD que representa la configuración inicial de un cuerpo. El cuerpo está sujeto a un cambio de configuración lineal (deformación) de la forma:

$$\begin{Bmatrix} x \\ y \end{Bmatrix} = \begin{Bmatrix} a_0 + a_1 X + a_2 Y \\ b_0 + b_1 X + b_2 Y \end{Bmatrix}$$

Tras la deformación, el punto A ocupa la posición a, B ocupa la posición b, C ocupa la posición c y D ocupa la posición d, como se indica en la figura.



- Calcule el campo de desplazamiento $\mathbf{u}(X, Y)$.
 - Calcule el campo tensorial de deformación de Green-Lagrange $\mathbf{E}(X, Y)$.
 - Calcule el campo tensorial de deformaciones infinitesimales $\boldsymbol{\varepsilon}(X, Y)$.
 - En el caso (c), calcule las deformaciones principales infinitesimales y direcciones principales para el punto de coordenadas $X = 0.5$; $Y = 0.5$.
2. La forma más general de un tensor isótropo de 4° rango es la siguiente:

$$C_{ijkl} = a_1 \delta_{ij} \delta_{kl} + a_2 (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + a_3 (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk})$$

donde a_1, a_2, a_3 son constantes reales.

- ¿Cuál es la forma que toma C_{ijkl} , para el caso del tensor de elasticidad de un sólido elástico lineal e isótropo?, ¿por qué?
 - Dé un significado a las constantes a_i que aparecen en la expresión del tensor de elasticidad de un sólido elástico lineal e isótropo.
3. Sea V el volumen encerrado por una superficie S de normal saliente unitaria $\mathbf{n}$, y sean $\phi = \phi(x_1, x_2, x_3)$ y $\psi = \psi(x_1, x_2, x_3)$ funciones escalares de las coordenadas x_i . Usando el teorema de Green, demostrar:

$$\int_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \mathbf{n} \, dS = \int_V (\phi \Delta \psi - \psi \Delta \phi) \, dV$$

donde $\Delta \psi = \psi_{,ii}$ es el laplaciano de la función ψ .

①

$$1) \quad a) \quad x = \alpha_0 + \alpha_1 X + \alpha_2 Y$$

$$y = \beta_0 + \beta_1 X + \beta_2 Y$$

$$x(0,0) = \alpha_0 = 0.1 \Rightarrow \alpha_0 = 0.1$$

$$y(0,0) = \beta_0 = 0.2 \Rightarrow \beta_0 = 0.2$$

$$x(1,0) = 0.1 + \alpha_1 = 1.2 \Rightarrow \alpha_1 = 1.1$$

$$x(0,1) = 0.1 + \alpha_2 = 0.4 \Rightarrow \alpha_2 = 0.3$$

$$y(1,0) = 0.2 + \beta_1 = 0.4 \Rightarrow \beta_1 = 0.2$$

$$y(0,1) = 0.2 + \beta_2 = 1.3 \Rightarrow \beta_2 = 1.1$$

Verif:

$$x(1,1) = 0.1 + 1.1 + 0.3 = 1.5 \quad \checkmark$$

$$y(1,1) = 0.2 + 0.2 + 1.1 = 1.5 \quad \checkmark$$

$$\underline{u} = \underline{x} - \underline{X} = \begin{pmatrix} 0.1 + 1.1X + 0.3Y - X \\ 0.2 + 0.2X + 1.1Y - Y \end{pmatrix} = \begin{pmatrix} 0.1 + 0.1X + 0.3Y \\ 0.2 + 0.2X + 0.1Y \end{pmatrix}$$

$$\underline{u} = \begin{pmatrix} 1 + X + 3Y \\ 2 + 2X + Y \end{pmatrix} \times \frac{1}{10}$$

$$b) \quad \underline{E}(X, Y) = ?$$

$$E_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} + \frac{\partial u_\alpha}{\partial X_i} \frac{\partial u_\alpha}{\partial X_j} \right)$$

$$\frac{\partial u_1}{\partial X_1} = \frac{1}{10} \quad \frac{\partial u_1}{\partial X_2} = \frac{3}{10} \quad \frac{\partial u_2}{\partial X_1} = \frac{2}{10} \quad \frac{\partial u_2}{\partial X_2} = \frac{1}{10}$$

(2)

$$E_{11} = \frac{1}{2} \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{10} \times \frac{1}{10} + \frac{2}{10} \times \frac{2}{10} \right) =$$

$$= \frac{1}{2} \frac{10 + 10 + 1 + 4}{100} = \frac{25}{200} = \frac{1}{8}$$

$$E_{22} = \frac{1}{2} \left(\frac{1}{10} + \frac{1}{10} + \frac{3}{10} \times \frac{3}{10} + \frac{1}{10} \times \frac{1}{10} \right) =$$

$$= \frac{1}{2} \left(\frac{10 + 10 + 9 + 1}{100} \right) = \frac{30}{200} = \frac{3}{20}$$

$$E_{12} = E_{21} = \frac{1}{2} \left(\frac{3}{10} + \frac{2}{10} + \frac{1}{10} \frac{3}{10} + \frac{2}{10} \frac{1}{10} \right) =$$

$$= \frac{1}{2} \left(\frac{30 + 20 + 3 + 2}{100} \right) = \frac{55}{200}$$

$$\underline{\underline{E}} = \begin{bmatrix} \frac{1}{8} & \frac{55}{200} \\ \frac{55}{200} & \frac{3}{20} \end{bmatrix} = \begin{bmatrix} 0.125 & 0.275 \\ 0.275 & 0.15 \end{bmatrix}$$

c) $\underline{\underline{e}}(X, Y) = ?$

$$e_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$e_{11} = \frac{1}{10}$$

$$e_{22} = \frac{1}{10}$$

$$e_{12} = e_{21} = \frac{5}{20} = \frac{1}{4}$$

$$\underline{\underline{e}} = \begin{bmatrix} 0.1 & 0.25 \\ 0.25 & 0.1 \end{bmatrix}$$

INDEPENDIENTE DEL PUNTO DE EVALUACION

(3)

$$d) \det \begin{pmatrix} 0.1-\lambda & 0.25 \\ 0.25 & 0.1-\lambda \end{pmatrix} = 0$$

$$0.1^2 - 2 \times \lambda \times 0.1 + \lambda^2 - 0.25^2 = 0$$

$$\frac{1}{100} - \frac{2}{10} \lambda + \lambda^2 - \frac{1}{16} = 0$$

$$\lambda^2 - \frac{2}{10} \lambda + \frac{16 - 100}{1600} = 0$$

$$\lambda^2 - \frac{2}{10} \lambda - \frac{84}{1600} = 0$$

$$1600\lambda^2 - 320\lambda - 84 = 0$$

$$\lambda_{1,2} = \frac{320 \pm \sqrt{320^2 + 4 \times 84 \times 1600}}{3200} = \frac{1}{10} \pm \frac{\sqrt{102400 + 537600}}{3200} = \frac{1}{10} \pm \frac{1}{4}$$

$$\lambda_1 = 0.35 \quad \lambda_2 = -0.15$$

$$\begin{pmatrix} -0.25 & 0.25 \\ 0.25 & -0.25 \end{pmatrix} \begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x'_1 = x'_2$$

$$\therefore (\lambda_1, \underline{x}') = \left(0.35, \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix} \right)$$

$$\begin{pmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{pmatrix} \begin{pmatrix} x''_1 \\ x''_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow x''_1 = -x''_2$$

$$v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2)

$$\textcircled{a} \quad C_{ijkl} = a_1 \delta_{ij} \delta_{kl} + a_2 (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + a_3 (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}) \quad \textcircled{1}$$

Para un material elástico :

$$\sigma_{ij} = C_{ijkl} \epsilon_{kl}$$

donde $\sigma_{ij} = \sigma_{ji}$ y $\epsilon_{kl} = \epsilon_{lk}$

Dada la simetría de $\underline{\sigma}$:

$$\sigma_{ij} = C_{ijkl} \epsilon_{kl} = \sigma_{ji} = C_{jikl} \epsilon_{kl} \quad \forall \epsilon_{kl}$$

Luego, debe ser

$$C_{ijkl} = C_{jikl}$$

En la expresión $\textcircled{1}$:

$$\textcircled{A} \quad C_{ijkl} = a_1 \delta_{ij} \delta_{kl} + a_2 (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + a_3 (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk})$$

$$\textcircled{B} \quad C_{jikl} = a_1 \delta_{ji} \delta_{kl} + a_2 (\delta_{jk} \delta_{il} + \delta_{jl} \delta_{ik}) + a_3 (\delta_{jk} \delta_{il} - \delta_{jl} \delta_{ik})$$

Hago (A) - (B) :

(5)

$$0 = 2 a_3 (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk})$$

$$\therefore \underline{a_3 = 0}$$

$$\therefore \underline{C_{ijkl} = a_1 \delta_{ij} \delta_{kl} + a_2 (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})}$$

b) Para un sólido lineal isótropo:

$$\sigma_{ij} = C_{ijkl} e_{kl} = \lambda e_{\alpha\alpha} \delta_{ij} + 2\mu e_{ij}$$

pero

$$C_{ijkl} e_{kl} = a_1 \delta_{ij} \delta_{kl} e_{kl} + a_2 (\delta_{ik} \delta_{jl} e_{kl} + \delta_{il} \delta_{jk} e_{kl}) =$$

$$= a_1 e_{kk} \delta_{ij} + a_2 e_{ij} + a_2 e_{ji} =$$

$$\stackrel{\uparrow}{=} a_1 e_{\alpha\alpha} \delta_{ij} + 2a_2 e_{ij}$$

Porque $e_{ij} = e_{ji}$

Luego

$$a_1 = \lambda$$

$$a_2 = \mu$$

(Ctes de Lamé)

$$\textcircled{A} \quad \int_S \phi \frac{\partial \psi}{\partial x_i} n_i dS = \int_V \frac{\partial}{\partial x_i} \left(\phi \frac{\partial \psi}{\partial x_i} \right) dV =$$

for
Theorem Gauss

$$= \int_V \left(\frac{\partial \phi}{\partial x_i} \frac{\partial \psi}{\partial x_i} + \phi \frac{\partial^2 \psi}{\partial x_i \partial x_i} \right) dV$$

$$\textcircled{B} \quad \int_S \psi \frac{\partial \phi}{\partial x_i} n_i dS = \int_V \left(\frac{\partial \psi}{\partial x_i} \frac{\partial \phi}{\partial x_i} + \psi \frac{\partial^2 \phi}{\partial x_i \partial x_i} \right) dV$$

$$\textcircled{A-B} \Rightarrow \int_S \left(\phi \frac{\partial \psi}{\partial x_i} - \psi \frac{\partial \phi}{\partial x_i} \right) n_i dS =$$

$$= \int_V \left(\cancel{\frac{\partial \phi}{\partial x_i} \frac{\partial \psi}{\partial x_i}} + \phi \frac{\partial^2 \psi}{\partial x_i \partial x_i} - \cancel{\frac{\partial \psi}{\partial x_i} \frac{\partial \phi}{\partial x_i}} - \psi \frac{\partial^2 \phi}{\partial x_i \partial x_i} \right) dV$$

$$\therefore \int_S \left(\phi \frac{\partial \psi}{\partial x_i} - \psi \frac{\partial \phi}{\partial x_i} \right) n_i dS = \int_V \left(\phi \frac{\partial^2 \psi}{\partial x_i \partial x_i} - \psi \frac{\partial^2 \phi}{\partial x_i \partial x_i} \right) dV$$
